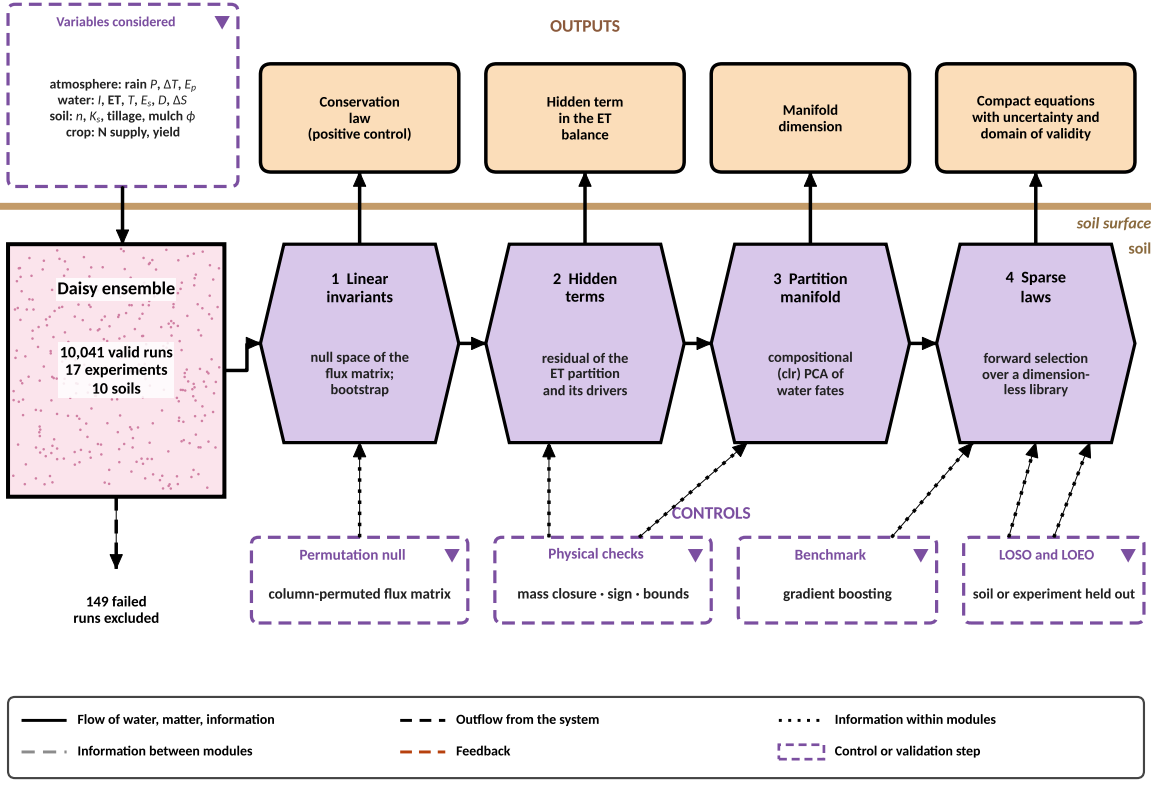

Variables considered
atmosphere: rain P, ΔT, E_p
water: I, ET, T, E_s, D, ΔS
soil: n, K_s, tillage, mulch ϕ
crop: N supply, yield
OUTPUTS
Conservation law (positive control)
Hidden term in the ET balance
Manifold dimension
Compact equations with uncertainty and domain of validity
soil surface
soil
Daisy ensemble
10,041 valid runs
17 experiments
10 soils
149 failed runs excluded
1 Linear invariants
null space of the flux matrix; bootstrap
2 Hidden terms
residual of the ET partition and its drivers
3 Partition manifold
compositional (clr) PCA of water fates
4 Sparse laws
forward selection over a dimension-less library
CONTROLS
Permutation null
column-permuted flux matrix
Physical checks
mass closure · sign · bounds
Benchmark
gradient boosting
LOSO and LOEO
soil or experiment held out
Flow of water, matter, information
Outflow from the system
Information within modules
Information between modules
Feedback
Control or validation step